

Course Outline for SFWR ENG 3O03 / COMP SCI 4O03 (6O03) Linear Optimization

McMaster University

Academic Year 2024-2025

Lectures/Instructor

Instructor: Anand Christopher

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Office: ETB 112

Lectures at LRW B1007:

- Monday, Thursday 12:30PM - 1:20PM
- Tuesday 1:30PM - 2:20PM
- Lectures will be broadcast on Teams.
- Some lectures will be delivered ONLY on Teams, due to health issues. Check Avenue for location before each class.

Office Hours: Tuesday 6:30 pm, on Teams

TAs/Tutorials

Teaching Assistants:

- Christopher Schankula
- Sheida Emdadi
- Tina Hatefisaeid
- Mary Emami
- Yimo Ning

Tutorial Sessions

- **T01:** Thursday 11:30AM - 12:20PM, JHE 326H
- **T02:** Friday 12:30PM - 1:20PM, ABB 271
- **T03:** Friday 9:30AM - 10:20AM, ABB 271
- **T04:** Thursday 11:30AM - 12:20PM, ABB 271

Final Grade Breakdown

Undergraduate Students:

- Assignments: 30% (3 x 10%)
- Midterm Exam: 30%
- Final Exam: 40%
- ***You must pass the final to pass the course!***

Graduate Students (6O03):

- Assignments: 20% (2 x 10%)
- Midterm Exam: 30%
- Final Exam: 30%
- Application Paper Presentation: 20%
- ***You must pass the final to pass the course!***

Course Description

Modelling and solutions for engineering and science problems using linear optimization, including networks, transportation, assignment, and scheduling problems. Solution methods include combinatorial algorithms such as simplex methods, primal-dual formulations, branch and bound formulations for mixed integer programming, and heuristics.

Slides, Assignments and Exams

- All course materials including slides, assignments, and exams will be made available through Avenue to Learn.
- The course schedule includes three written assignments, one midterm, and one final exam.
- Please consult Avenue to Learn for the specific handout dates and due dates for assignments, as well as the scheduled dates for the midterm and final exams.
- In cases of deferred exams, the instructor reserves the right to conduct them orally.

Course Policies and Notices

- **Assignment Submission:** All solutions must be submitted to GitHub classroom by the due date. You will find a link to the assignment templates on Avenue. Late submissions will incur a penalty of 20% per day.
- **Integrity in Submissions:** You must commit partial work to your GitHub repo every 10 minutes. Students must fill out the LICENSE.txt in each assignment repo. References must be provided for any external sources used.
- **MSAF Policy:** If an assignment is missed due to a valid reason, the weight may be transferred to the final exam under the MSAF policy.
- **Grading Concerns:** Any complaints regarding grading must be addressed within two weeks of the date the work was returned.
- **Loss of Work:** The instructor and TAs assume no responsibility for lost assignments or exams.

Suggested Textbooks

- “Introduction to Mathematical Optimization”, Matteo Fischetti. (cheap)
- “Linear and Nonlinear Optimization”
- “Operations Research: Deterministic Optimization Models” by Katta G. Murty, Prentice Hall, 1995

Course Overview

Topics covered in the course include:

- Introduction to Linear Optimization
- Mathematical Models for Linear Optimization
- Geometric Analysis of Linear Optimization
- Simplex Method
- Dynamic Programming
- Combinatorial Optimization on Graphs
- Network Flow
- Integer Programming
- Applications

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty. Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/> The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism Detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g., Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty. Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster’s use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Generative AI

Students may freely use generative AI in this course so long as the use of generative AI is referenced and cited including exact quotations for all prompts used. Use of generative AI without citation will constitute academic dishonesty. It is the student’s responsibility to be clear on the expectations for citation and reference and to do so appropriately.

Courses with an Online Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program

affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online. It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s Academic Accommodation of Students with Disabilities policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the Policy on Requests for Relief for Missed Academic Term Work.

Academic Accommodation for Religious, Indigenous or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.